THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The correct name of the covalent compound PCl_3 is: 1

(a) phosphoric chloride **(b)** phosphorus trichloride
(c) phosphorus chloride **(d)** monophosphorus trichloride

2. Which postulate of Dalton's atomic theory is correct? 1

(a) Atoms cannot be created or destroyed in a chemical reaction
(b) Atoms of different elements are identical
(c) Atoms can be divided in a chemical reaction
(d) Atoms combine in any ratio

3. In a nitrogen molecule the number of shared electron pairs is: 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

4. Which compound is ionic? 1

(a) $MgCl_2$ **(b)** HCl **(c)** CO_2 **(d)** CH_4

5. Sugar solution does not conduct electricity because: 1

(a) water is a conductor only **(b)** sugar forms Na^+ ions
(c) sugar is covalent and does not form ions **(d)** sugar is a metal

6. In Thomson's plum pudding model the electrons are: 1

(a) orbiting a nucleus **(b)** embedded in a sphere of positive charge
(c) inside the nucleus **(d)** absent

7. Rutherford's model could NOT explain: 1

(a) the stability of the atom **(b)** scattering of α-particles
(c) the existence of a nucleus **(d)** the neutrality of atoms

8. Which subatomic particle is neutral and found in the nucleus? 1

(a) electron **(b)** proton **(c)** cation **(d)** neutron

9. Atoms with the same mass number but different atomic numbers are called: 1

(a) isotopes **(b)** isomers **(c)** isobars **(d)** isotones only

10. The weighted average atomic mass of chlorine (35 u : 37 u = 3 : 1) is: 1

(a) 37 u **(b)** 35.5 u **(c)** 35 u **(d)** 36 u

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the compounds: 2

(a) CO
(b) N_2O_5
(c) H_2S
(d) SF_6.

12. How does sodium chloride form? Write the ions. 2

13. State the rules for filling of electrons in shells (Bohr–Bury). 2

14. Write the electronic configuration of sodium (Z = 11) and chlorine (Z = 17). Give the valency of each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Write the formulae of 3
(i) aluminium oxide
(ii) calcium carbonate
(iii) sodium sulfate using the criss-cross method for one of them.

16. Distinguish between isotopes and isobars with one example each. 3

17. Write the names of: 3
(a) CS_2
(b) N_2O_4
(c) PCl_3
(d) H_2S
(e) NH_3 (common name). State the rule for prefixes.

OR

Distinguish between ionic and covalent compounds on the basis of 3
(i) bonding
(ii) melting point
(iii) conduction of electricity.

18. Name the three isotopes of hydrogen and give the number of protons and neutrons in each. 3

OR

State three uses of isotopes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Compare ionic and covalent compounds in four points. 5
(ii) Why do ionic compounds dissolve in water while many covalent compounds do not?

OR

Case study: A chemist records the substances CO_2, NH_3, NaCl and $MgCl_2$. 5
(i) Classify each as molecular (covalent) or ionic.
(ii) Find the molecular mass of CO_2 and NH_3.
(iii) Find the formula unit mass of NaCl and $MgCl_2$.
(iv) Why are formula unit masses used for ionic compounds? (H=1, C=12, N=14, O=16, Na=23, Mg=24, Cl=35.5)

20. (i) Define isobars and give an example. 5
(ii) Calcium (Z = 20), potassium (Z = 19) and argon (Z = 18) each have mass number 40. Find the neutrons in each.
(iii) Distinguish isotopes from isobars.

OR

(i) State five uses of isotopes. 5
(ii) Why can one not use ordinary chlorine for the same purposes?
(iii) Name one Indian scientist associated with nuclear energy.

END OF QUESTION PAPER — **Total: 40 Marks**